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The Political Economy and Ecology of Capture Fisheries: Market Dynamics, Resource Access and Relations of Exploitation and Resistance

LIAM CAMPLING, ELIZABETH HAVICE AND PENNY McCALL HOWARD

Capture fisheries are constituted through historically specific environmental conditions and social and economic relations of production. Fisheries, whether saltwater or freshwater, are an important source of animal protein, livelihoods and exchange value in international trade, and are presently undergoing rapid socio-ecological change. To explore the political economy and ecology of capture fisheries around the world, this paper synthesizes the insights of 11 empirical studies and places fisheries in the broader context of the capitalist relations of production through which they operate. The competitive market dynamics of fisheries production and consumption are examined, as well as the forms of social-property relations, social differentiation, labour exploitation and resistance that occur within them. This paper highlights some of the ways in which the unique combination of characteristics associated with fish and fisheries complement and complicate familiar questions in agrarian political economy. It concludes by identifying future research directions.

Keywords: fish, fisheries, resource access, political economy, state, capitalism

INTRODUCTION

With this special issue, the *Journal of Agrarian Change* makes a foray into expanding its analysis of the 'social relations and dynamics of production, property and power in agrarian formations and their processes of change' from land into aquatic environments. Fish, whether saltwater or freshwater, farmed or captured, are an important source of both food and income and are noted as an area of 'fundamental significance' to agrarian political economy in the inaugural essay of this *Journal* (Bernstein and Byres 2001, 37).[1] Three-fourths of the world's fisheries are at or beyond 'full exploitation', indicating the likelihood that many fish populations, and the ecosystem of which they are a part, will decline (if they are not already) with current and expanded levels of

Liam Campling, School of Business and Management, Queen Mary, University of London, Mile End Road, London E1 4NS, UK. E-mail: l.campling@qmul.ac.uk; Elizabeth Havice, Department of Geography, University of North Carolina-Chapel Hill, 324 Saunders Hall, Campus Box 3220, Chapel Hill, NC 27599 USA. E-mail: havice@email.unc.edu; Penny McCall Howard, Level 2, 365 Sussex Street, Sydney, NSW 2000, Australia. E-mail: suilven2@yahoo.com. Authorship of this paper and editorial collaboration to produce this special issue was fully collective from start to finish.

We express huge thanks to the authors for their commitment and contribution, and to the four editors of this *Journal* for supporting this project and for commenting constructively on post-peer review drafts of all papers featured here. We also heartily thank the 32 peer reviewers for their time and insights, Fiona McCall for her eagle-eyed copy editing of several papers, and Deborah Johnston, Cristóbal Kay and Barbara Neis for their comments on an earlier draft of this paper.

[1] Relatively few research papers on fisheries have featured in this *Journal*. Since its launch in 2001, only five have been published: three on aquaculture systems (Hall 2004; Belton and Little 2008; Adduci 2009) and two on capture fisheries (Neiland et al. 2005; Dressler and Fabinyi 2011).

competitive extraction. This is cause for concern because fish, shellfish and marine mammals (hereon 'fish') are an important source of animal protein, hundreds of millions of people are employed as fish workers and in fisheries-related activities, and fish exports from developing countries generate a higher export value than coffee, bananas, cocoa, tea, sugar and tobacco *combined* (FAO 2010). Despite the social importance of fisheries for food and income, fisheries researchers have expressed concerns that 'social objectives' in fisheries policy have been 'subsumed under the goals of economic growth and wealth creation' (Symes and Phillipson 2009, 1).

In this special issue, we seek to put analyses of capture fisheries in the broader context of the capitalist relations of production through which most now operate. Mainstream social science work on fisheries often avoids, if not outrightly obscures, how capitalist relations and dynamics (in their diverse and varying forms) shape fisheries systems. The prevailing treatment of fisheries in the social sciences has been biologically and economically reductionist. Fisheries have been treated as 'a technicality, an exercise of narrow, instrumental rationality ruled by universal theory' (Jentoft 2007, 435). Further, within this tradition, rational choice theory or models 'depict the individual producer as an autonomous isolate engaged in the technical act of catching fish' (Pálsson 1991, 21). Debates in the few social science journals devoted to marine issues, such as *Marine Policy* and *Maritime Studies* (*MAST*), are framed primarily through a policy-orientated lens, where (often undifferentiated) fishing 'communities' are frequently cited as subjects in and for the better 'management' of resources.

Outside of this dominate framing, a corpus of social scientists have generated a robust, but small, critical literature investigating fisheries from the perspectives of, *inter alia*, feminist and labour-centred anthropology and sociology (e.g. Kurien 1978; Acheson 1988; Muszynski 1996; Nadel-Klein and Davis 1988; Davis 1996; Menzies 2002; Sider 2003; Neis et al. 2005), and geographies of state and capital in fisheries property systems (e.g. Steinberg 2001; Bradshaw 2004; Mansfield 2004a,b, 2007; Sneddon 2007; Fougères 2008).[2] In this vein, the 11 papers in this issue demonstrate the historical and socio-ecological complexity of capture fisheries. In this introductory article, we draw on their empirical findings and analytical contributions, systematically distilling themes that are significant in the study of the political economy and ecology of capture fisheries systems and to readers of this *Journal*.[3]

In capture fisheries, fish are hunted in wild marine or freshwater ecosystems using techniques ranging from spears, traps and hooks, to massive nets guided by sophisticated fish-finding sonar technologies. As a source of food, in 2007, fish accounted for 15.7 per cent of the global population's intake of animal protein, and the per capita supply of fish for food has skyrocketed from less than 3 kilograms in 1950, to 17.2 kilograms in 2009. Capture fisheries accounted for around 62 per cent of total fish production for 'all uses' in 2009; aquaculture systems supplied the remaining 38 per cent (FAO 2010).[4] Unpacking the term 'all uses' reveals that fish are used

[2] This is not a comprehensive list. The bibliographies of the papers in this issue provide a wealth of resources for those interested in a political economy of fisheries.

[3] This special issue excludes aquaculture. The World Bank distinguishes between capture fisheries where 'fish are captured through the use of gear' and aquaculture where 'fish are farmed or raised' (World Bank 2004, x). To an extent, this distinction obscures commercial realities because fish farming often relies on wild-caught fish as feed or 'fish meal', which is processed into an industrial feed product either from whole fish (e.g. anchovy) or from by-products of fish processing for human consumption (e.g. parts of fish that are not suitable for human consumption) (see e.g. Naylor et al. 2009). Some aquaculture facilities grow/'fatten' wild-caught fish (Longo and Clark 2012), and some are devoted to raising juvenile fish to be released into the wild to 'stock' populations (Naylor et al. 2007). Aquaculture has been subject to considerable interrogation in the critical social sciences (e.g. Vandergeest et al. 1999; Goss et al. 2000; Stonich and Bailey 2000; Stonich and Vandergeest 2001).

[4] World marine and inland capture fisheries catch volume increased by as much as 6 per cent per year in the 1950s and 1960, trebling from 18 million tonnes in 1950 to 56 million tonnes in 1969. During the 1970s and 1980s, the average rate of increase declined to 2 per cent per year and fell to zero, and declined after the 1990s

for human consumption as well as non-food uses, such as fertilizers and 'industrial' fish meal for agricultural feed (e.g. chickens, which in turn are for human consumption).

Capture fisheries are not only a source of use values. In international exchange, fish had a *first-sale* value of US$93.9 billion in 2008 (FAO 2010). The World Bank estimates that 'about 120 million people are directly dependent on commercial capture fisheries for their livelihoods as fulltime or part time workers, including employment in the post-harvest sector' (World Bank 2010, 4). Behind these generalized headline figures are highly differentiated capture fisheries systems at scales that vary from subsistence (that supports day-to-day consumption) to industrial commodity production (where returns are sufficient to merit investment from equity funds).

Capture fisheries have several peculiar features, the *combination* of which make them analytically unique and intellectually challenging. Fish are the last hunted commodity on the planet and are a resource that is renewable, but exhaustible. Defining, allocating, owning or managing rights to extract particular fish species in specified areas is complex because fish live in water and are mobile. Ecology, market dynamics, law and geopolitics have shaped social–property relations in fisheries in various historically contingent ways.[5] As a result, fisheries property relations have taken many forms (e.g. Hollick 1981; McEvoy 1986; Ostrom 1990; Pontecorvo 1988; Schlager and Ostrom 1992; Dodds 2000; Steinberg 2001).

In the contemporary era, states are the primary arbiter of property relations in most inland and marine capture fisheries.[6] States regulate fishing activities in many ways, such as by time, technology, technique, catch volume and/or space. For example, to limit fishing time, fisheries management agencies may designate a total number of fishing days or establish a fishing season. To regulate technology or technique, a fishery may be restricted to particular vessels based on their size or engine power, gear type (e.g. nets versus hooks) and/or the specific attributes of fishing gear (e.g. the mesh size of nets). To limit the spatial range of activities of fishing interests, states may open or close geographical areas to fishing. State sovereignty over the 200 nautical mile exclusive economic zone (EEZ) is the most prominent example of spatial regulation of fishing. Sovereignty over the EEZ emerged in the mid-1970s in customary international law and enabled states to regulate fishing activity and to determine which actors access marine resources. In 1982, the United Nations Convention on the Law of the Sea (UNCLOS 1982) enshrined EEZs in international law.[7] Recently, spatial regulation in the oceans has taken the form of marine protected areas – where managers can restrict fishing activities in ecologically important areas (e.g. fish spawning grounds) – and marine spatial planning – where marine activities, including fishing, are designated to defined ocean spaces.

(FAO 2000). Catch volume was steady over the period from 2006 to 2008 at about 89.8 million tonnes (FAO 2010).

5 We use social–property relations following Brenner (1986) and interchangeably with 'property rights'.

6 For historical overviews of states and fisheries regulation from a variety of methodological perspectives, see Bavington (2010), Hannesson (2004), Roberts (2007) and Steinberg (2001).

7 Social–property relations over the ocean have undergone a transformation over the past 400 years from the conception of the ocean (and the resources there within) as 'open access' to the recognition of state sovereignty over coastal zones. In the early 1600s, Dutch jurist Hugo Grotius (1916) argued for the freedom of the seas (*mare liberum*), where ocean navigation and fisheries resources are open to all. At the height of Dutch maritime power, Grotius was 'alarmed that a few nations might try to gain control of shipping lanes and restrict the advantageous commerce of his native land' (Butler 1990; Bromley 2008, 9). Freedom of the seas was a call to prevent national claims to political control or territorial sovereignty over the oceans, apart from a slim strip of coastal waters. This doctrine largely held for the next 350 years until 1945, when the United States extended national jurisdiction 'to all resources on the subsoil and seabed of the continental shelf'; a move motivated by interest in mineral resources (mainly oil) (Cantorna et al. 2007, 373 n35). In 1947, Chile, Ecuador and Peru lead the shift towards ocean sovereignty with similar declarations (Loftas 1981).

Today, the individual transferable quota (ITQ) is the most common approach to fisheries management. ITQs purport to overcome the challenges associated with regulating mobile fish resources by establish individual rights to specific quantities of fisheries resources (see below on the 'tragedy of the commons'). Upon defining a quantity of fishing rights, states allocate this quantity to interests in the form of quota. Quota holders are permitted to trade their share. The effect is that quotas acquire a commodity value, which in turn, generates struggles among actors to appropriate this value (see e.g. Mansfield 2004a). In general, fisheries access rights are difficult to monitor and enforce, and are politically contested (see below on 'Resource Access and States').

Science plays multiple roles in fisheries systems. It is used to identify commercially viable fisheries; develop techniques of extraction and post-catch handling and distribution; and, assess the status and health of the natural resource. Science is also used to mediate ambiguous property relations because it informs 'management' decisions (including the definition and allocation of fishing rights). Fisheries scientists work in universities, specialized government research laboratories, and directly for government fisheries departments. Like all science, fisheries science is plagued with uncertainty. In this case, uncertainty arises at least in part from the practical difficulty of estimating populations of uncaught fish swimming in an ocean, lake or river; accounting for ecosystem fluctuations; working with often incomplete data on fishing vessel catches; and, on this basis, trying to predict future fish stocks and mediate often contending political and economic interests (on these topics, see e.g. Botsford et al. 1997; Beddington et al. 2007).

In the context of competitive fisheries extraction, science-based management generally aims to find the maximum 'sustainable yield' that a particular fishery can support without its population declining. In cases when fishing pressure goes beyond this point of maximum 'efficiency', management is also charged with correcting, mitigating or diminishing the environmental consequences of increased demand for limited fisheries resources. Fisheries policy more broadly is commonly technical in formulation and relies on management recommendations generated by bio-economic management models, continuous improvements in fisheries science and/or associated data (see e.g. Cochrane 2002). Fisheries policy is also used to target particular practices and policies that contribute to fisheries decline, such as fisheries subsidies (Campling and Havice 2010; von Moltke 2011).[8] New management tools have also been deployed, including ecosystem-based management (Pikitch et al. 2004), market-based mechanisms that identify and purportedly reward 'sustainable fisheries' in the marketplace (Gulbrandsen 2009), and methods for integrating fishers ecological knowledge and fishers themselves into management (Neis and Felt 2000; Neiland et al. 2005).

Towards a Political Economy and Ecology of Capture Fisheries

Global attention has turned towards the widely reported state of crisis in capture fisheries. Scientific studies detail fish and ecosystem decline, predict collapse and debate current fish population status and management efficacy (Jackson et al. 2001; Myers and Worm 2003; Worm et al. 2006; Mora et al. 2009; Worm et al. 2009). Meanwhile, popular literature relates stories of dramatic fisheries collapse, for example of the iconic Newfoundland cod fisheries in the mid-1990s, and the plight of the majestic Atlantic bluefin unfolding today (Kurlansky 1998; Clover 2006, Greenberg 2010). These examples reflect a more general trend towards steady

[8] Notably, international efforts to eliminate fisheries subsidies contradict with policies from within states to use subsidies to support fisheries economically.

geographical extension and socio-technical intensification of fisheries extraction and draw broad attention to the future of fish populations (as well as the food security and economic activities that they support).

There are many theoretical debates around these issues. Particularly resonant, and familiar to readers of this *Journal*, are neo-Malthusian, neo-populist and Marxist theorizing on nature–society relations. In fact, the analysis of fisheries systems and their decline are foundational to neo-Malthusian arguments that human population pressures yield the 'tragedy of the commons' (Gordon 1954; Hardin 1968). In this view, fisheries exemplify how, in a world with a growing population, but without private property regimes, natural resources are destined to decline and become more scarce. These works have shaped neo-Malthusian theorizing generally, and laid the foundations for modern fisheries management. From this perspective, establishing property relations in fisheries will overcome the market failures associated with open access regimes (e.g. the negative environmental outcomes of competing fishing interests). Scott explicitly identifies property relations as *political economy* considerations and argues that 'the social optimum in the long run and the short run would demand that common-property resources be allocated to maximizing owners' (1955, 124).[9] In other words, Scott reduces political economy to the mechanism of sole ownership of fisheries resources and suggests that ownership generates vested interests with a stake in the economic sustainability of the fishery. This is a narrow definition of political economy dynamics in fisheries, not least because, as Ostrom (1990) points out, the direct subject of fisheries management is not fish, but people who are embedded in existing social, political and economic institutions. Further, assuming that private property relations will 'appropriately' distribute resources according to a single efficiency maximizing equation treats capital as a 'thing' rather than examining the diverse social relations that constitute it (Durrenberger 1997).

There are also familiar neo-populist themes in the fisheries literature. Echoing Schumacher's (1973) *Small is Beautiful*, contemporary debates over fisheries sustainability and food sovereignty often idealize small fishing boats as environmentally and socially beneficial, in contrast to highly capitalized industrial fisheries which are depicted as driving fisheries depletion and being controlled by greedy corporations. This position is particularly pronounced in the advocacy of environmental NGOs and in market-based sustainability mechanisms. An example from tuna fisheries draws out this point. Greenpeace (2009) argues that replacing large, capital-intensive industrial tuna fishing boats with smaller, labour-intensive vessels will improve environmental conditions and create development opportunities. Likewise, UK supermarket retailers and other major buyers of fish products have incorporated 'small is beautiful' into their 'sustainable' procurement strategies, committing to purchase, for example, only canned tuna products caught using smaller scale fishing boats (Hamilton et al. 2011). Offering an image of small scale, sustainable fisheries production that is in harmony with the environment is a useful campaigning, advocacy and marketing tool. Switching to smaller scale fishing may indeed generate stated environmental and economic goals for particular fisheries. However, this neo-populist worldview can obscure the reality that vessel size and the scale of production alone do not determine the social and/or environmental relations and effects related to fisheries systems.

Finally, as with research on agri-business, the fisheries literature contains interventions from a Marxian political economy perspective. This work applies Marxist concepts to offer broad-

[9] Hardin's (1974) commentary echos and reiterates this sentiment by arguing that because resources will be insufficient to support population increases, they should be allocated to the more productive members of global society.

brush accounts of capitalist crisis in fisheries and sketches the tendency of capital in fisheries to undermine its ecological bases of production (e.g. Clausen and Clark 2005; Clark and Clausen 2008; Longo and Clausen 2011; Skaladany et al. 2005). These interventions highlight the contradictions of capitalism in fisheries on a global scale, but can obscure historical and scientific specificity and socio-ecological diversity in the world's fisheries. Several contributions to this issue provide materialist analyses of fisheries, offering a bridge between the abstract categories of Marxist political economy and 'a concrete analysis of a concrete situation' (Lenin 1965, 166) in particular fisheries.

From these points of departure, the papers in this issue investigate how capitalist relations and dynamics (in their diverse and varying forms) shape and/or constitute fisheries systems. They reveal how we get fish out of the water as food and commodities, and to what socio-ecological effect. The papers in this issue are organized in three parts: market dynamics and competition in fisheries production-consumption systems; labour and relations of exploitation and resistance; and resource access and states. We use the remainder of this introductory paper to contextualize these three themes and put the 11 studies in this issue in direct conversation with each other. We conclude by identifying gaps not sufficiently addressed in this issue as well as research directions that can further illuminate the political economy and ecology of capture fisheries. All authors cited herein without a year reference are papers in this special issue (2012).

MARKET DYNAMICS AND COMPETITION IN FISHERIES PRODUCTION-CONSUMPTION SYSTEMS

This theme enables an investigation of how capitalist development copes with the challenges of extracting fish and bringing them, profitably, to market. Integral is how the social relations of capitalism constitute historical trajectories of environmental and social change and contemporary dynamics in fisheries production systems. The five papers that address these issues head-on are grouped in *Part I* of this special issue. They bring to life a range of market dynamics in fisheries, including for high-value sashimi tuna in the Mediterranean (Longo and Clark); household shrimp production in Louisiana, USA (Marks); low-value canned tuna in the Western Indian Ocean (Campling); artisanal and small(er)-scale fisheries in the riverine Mekong ecosystem (Sneddon and Fox); as well as the interests seeking to make markets for 'sustainable' fish (Ponte). In addition to these five papers, we highlight where other contributions to this issue reveal competitive pressures at and between the point of extraction and the world-market.

Transitions to capitalism mean that, in most cases, fish have changed from being produced as food for producing communities to being produced as commodities for sale. This is a classic concern of agrarian political economy, including the subsequent effects of 'the commodification of subsistence' on all aspects of social relations (Bernstein 2010a, 65). Capitalism works through nature at sea in the particular geographical grounds where fish are present, in the techniques developed to capture them, and in the organizational strategies of firms that operate fishing vessels. But at the same time, capitalist production has a tendency to undermine the environmental conditions for its own reproduction: for example, this drive to create profit has undermined several tuna fisheries and tuna populations (Campling; Longo and Clark). In two different North American shrimp fisheries, Marks and Foley each examine how change and crises in commodity and price relations create both new opportunities and sometimes devastating pressures on fishing and processing firms.

State and market pressures from outside the fishing industry can also shape the ecological resources that fisheries depend upon. For example, building hydrodams on the Mekong for industrial development purposes will destroy and/or displace fisheries production systems.

The political project of developing hydropower infrastructure involves assessing the perceived value of fisheries systems in relation to the predicted value of hydropower (Sneddon and Fox). Processes of commodification continue to develop in innovative ways, as in the rise of retail-centred standards for fish where 'sustainability' becomes a commodity (Ponte). From here, we highlight a series of sub-themes that further illustrate the ways in which market dynamics and commodity relations constitute contemporary fisheries, in particular: the role of firms and commodity chains; different techniques of production; the environmental conditions of production; consumption; and the politics of fisheries and development in the 'global South'.

Firms and Commodity Chains

In the mainstream fisheries management literature, boats are abstract units of production. Fishing vessels are differentiated (if at all) by vessel size, fishing gear, and country of registry (also referred to as 'flag', the country to which the vessel is legally responsible). This relatively simplistic framing obscures who owns and controls fishing vessels and their relations with each other and with other firms in fisheries commodity chains. Further, it hides the fact that many vessel owners purposely obscure ownership and separate vessel ownership and registration to evade regulation and reduce costs.[10]

Several papers in this special issue focus on the nature and role of firms in fisheries production systems. Campling, Howard and Marks each show that there are a diversity of types of fishing firms and accumulation strategies in fisheries, ranging from multinational corporations through to household producers and individual owner-operators. Throughout this special issue, we see examples of varying levels of concentration and control, ownership structure, and models of industrial organization. This includes relationships between fishing firms and interests in other functional activities in the commodity chain, particularly processing (the manufacturing node of fisheries production) and trading companies (see Foley; Havice and Reed). In other words, despite appearing to engage in the same functional activity, the composition and relations of the firms that control fishing boats are varied. No single model of production explains the organization of fishing and associated activities.

In the context of the world market, fishing firms compete both horizontally (with other boats producing the same commodity, although often in separate, widely dispersed fisheries) and vertically in different functional activities (e.g. when they negotiate over the price of their product with fish traders and processors). Foley, Marks and Sinha provide distinct accounts of shrimp producers struggling to remain competitive in different parts of the world. Though not explicitly elaborated in the papers, when read together, they graphically illustrate the competitive dynamics of producing for and in the world market. Indirectly, each fishery experiences pressure from and exerts pressure on the other. Drawing on Chayanov (1991), Marks shows how shrimpers engaged in household production in Louisiana deepen their 'self-exploitation' in order to survive heightening competition from export-orientated producers around the

[10] One tactic to do so is the use of 'flags of convenience' where the vessels of 'national' firms chose not to register under the flag of their country of ownership. Vessels are mobile and many operate in the waters of multiple countries. They are required to abide by laws in the waters of the countries in which they are fishing as well as to their 'flag state' (the state in which the vessel is registered) for regulatory issues (e.g. labour and environment) (e.g. DeSombre 2006). Vessels are also inspected at port by government agencies (port state control) to ensure they meet regionally agreed standards, although enforcement is focused on larger cargo vessels rather than smaller fishing vessels. Without access to expensive commercial databases such as IHS Fairplay's 'Sea-web', gathering even simple data on vessels can be difficult since ownership and registration are frequently changed. The Environmental Justice Foundation (2010) argues that the flag of convenience systems facilitates illegal fishing.

world. Sinha shows the huge importance of industrial export-orientated shrimp production to the trajectory of fisheries development in Kerala and its associated social conflicts over time (see below on Foley).

Longo and Clark detail horizontal competition between interests that use two distinct gear types to exploit the same population of bluefin tuna in the Mediterranean. Under changing world market conditions, modern, highly capitalized purse-seine vessels have superseded the fixed nets systems (*tonnare*) which were developed as early as the second century CE. Foley shows how firm-level control over 'sustainability' labelling of shrimp has emerged as a tool of accumulation and generated both inter-capitalist co-operation (horizontally between processors) and competition (both among processors and between processors and other fishing interests). His article demonstrates how firms in Newfoundland, Canada collaborate and compete to secure raw material supply, and use the Marine Stewardship ecolabel to do so.

No author in this issue traces a fisheries commodity chain from the point of production to its eventual consumption, and this is an area that requires attention in future studies of the political economy and ecology of capture fisheries.[11] Papers in this issue reveal themes that feature in many fisheries commodity chains. As in other contemporary food chains, world-market cost structures and emergent standards shape power relations in fisheries production systems (Foley; Havice and Reed; Howard; Longo and Clark; Marks; Ponte). In addition, understanding how environmental conditions and seasonality shape the business strategies of firms as they transform nature into commodities (Campling; Howse et al.) is central to the organization of *fisheries* commodity chains.

Techniques of Production

Ships are machines operated by labourers whose co-operation and collectivity is crucial to the survival of the enterprise and of workers themselves (Rediker 1989). Where fishing boats are machines designed to produce value through capitalist relations of production, the valorization process has a significant influence on the selection, design and development of these machines (MacKenzie 1996). Yet 'instruments' are only one part of the labour process in fisheries systems, so we use the term 'techniques' to include the labour of operating technologies (e.g. fishing gear and vessels), and the other social relations this entails, such as disciplinary managerial regimes. The ability of owners to increase the level of capital investment in boats and expand their fleets is predicated on the sustained extraction of value through the fishing labour process. Surplus value extracted elsewhere in the commodity chain, for example by fish processors, may be used to fund backward integration into fishing (Campling). Fishing-related investments may not directly support resource extraction, but may also be made to mitigate risk and ensure conditions for future production of surplus value. For example, Havice and Reed show that fishing and trading firms invest in tuna processing plants in Papua New Guinea not to profit from processing activities, but because the government offers long-term fishing licences in exchange for the investment.

The development of specific techniques of fisheries production used in particular fisheries is shaped by existing environmental conditions, available boats and gear, market pressures (and opportunities), transport infrastructure, and social relations in and around fisheries systems. Constant change and innovation in all aspects of fishing, transportation, processing and mar-

[11] See Gibbon (1997) for a detailed study of prawn commodity chains centred on production in Tanzania. Wilkinson (2006) offers an account of a generic commodity chain in 'fish'. See also the special issue of this *Journal* on global commodity chains and African export agriculture (Daviron and Gibbon 2002).

keting techniques is a feature of commercial fisheries, as boat owners and processors seek to catch fish and get them to market in the most economically efficient manner. Markets are often distant from the location of extraction. As a result, firms have had to overcome the social and technical constraints of geographical distance to market and the organic durability of highly perishable fish products (e.g. Campling; Friedmann 1992). Firms deploy particular techniques in response to different market demands and cultures of consumption. For example, in the low-value canned tuna sector, firms use techniques that maximize catch volume and are less sensitive to quality differentials (Campling). In contrast, to access the extremely high-value, quality-sensitive tuna sashimi market, firms developed techniques that maintain and enhance quality, for example by fattening wild-caught fish in pens in the ocean (a technique known as tuna 'ranching') (Longo and Clark).

Conflicts between fishers and fishing firms using different gear types and fishing techniques are common. Such 'gear conflicts' have a social and/or economic basis and impact (e.g. Phyne 1990). This is illustrated in Kerala where 'small scale' fishers organized politically to improve the efficiency of their own techniques and to strengthen their position *vis-à-vis* industrial, more capital-intensive fishing gears (Sinha). In Scotland, there is conflict between vessels that fish the same grounds, but use different fishing gears to produce nephrops/langoustine for distinct (high and low value) market segments (Howard). Owners' choice of techniques can have substantial social impacts, for example, the design of fish processing plants can have a considerable effect on the likelihood that workers will develop occupational diseases that can affect the rest of their lives (Howse et al.).

The Environmental Conditions of Production

Existing ecological conditions form the basis for different fishing techniques and the social relations involved in production and exchange in fisheries. Sneddon and Fox detail the dramatic ecological characteristics of the Mekong River delta flood cycle that have given life to one of the most productive and diverse freshwater fisheries in the world. The material pulse of the flood cycle is also a potential resource for hydro-power development projects that would capture and alter the river flow that supports fisheries. Sinha highlights the environmental characteristics that divide Kerala's coast into three distinct fishing zones. Each zone presents ecological conditions that are 'home' to specific fish species; in turn, fishers have adapted fishing gears and techniques to the target species and sea conditions (e.g. sheltered versus exposed coastlines that shelve steeply or gradually into the deep water of the Indian Ocean basin). Notably, Kerala's dual monsoon makes waters too rough for small craft for several weeks of the year. On the one hand, rough seas are a problem for small producers because they generate seasonal income dips. On the other hand, many fish varieties breed during the rough sea season and reduced fishing pressure during this time better enables fish populations to regenerate.

Papers by Campling and Longo and Clark highlight how the biology of various tuna species make them more resilient or vulnerable to industrial-scale fishing pressure. Bluefin and yellowfin tuna are slow-growing and late to sexual maturity and as a result are more negatively affected by industrial scale fishing than skipjack, which grow quickly and reach sexual maturity years before their larger (and also more commercially valuable) brethren. Both papers explore how industry has adapted to, or ignored, these biological features, and to what effect, offering concrete examples of the ways that capitalism develops through, and produces nature, often with unintended (but not unsurprising) ecological consequences (see, e.g., Smith 2008 [1984]; Moore 2010a,b).

Fuzzy consumer ideals about the appropriate use of aquatic resources reflect popular conceptions of the sea as a wilderness that should be preserved from human influence. Like the land, the sea cannot produce food for billions of people, generate income for millions, *and* remain in a 'wilderness' state. Rapid growth in ecolabel schemes reflects a general, but abstract, demand for fish production systems that feed humanity without causing environmental harm. Ponte reveals that, so far, seafood ecolabels do not fulfil this dual mandate. Virtually all sustainable fishing schemes and fisheries management policies are framed with 'the normative subject "fishermen exploiting the seas"', and implicitly require that fishermen accept that their activities are destructive (Nightingale 2011, 6). This is in contrast to farming, where the effects of human labour in the environment are more readily seen as productive.

Economies and livelihoods that depend on single-species commercial fisheries have been devastated when the target species collapses, but environmental catastrophe can lead to new opportunities for some firms. After cod populations collapsed in Newfoundland, their prey, shrimp, flourished, and eventually became the basis for a new, valuable fishery (Foley). The dramatic cod collapse and subsequent criticism of state fisheries management also established the political conditions for the establishment and spread of private ecolabelling schemes. In turn, ecolabel schemes created a new market for 'sustainable' seafood and a lucrative opportunity for those firms able to access and capitalize upon such schemes (Foley; Ponte). Firms innovate and adapt to stave off economic crises associated with environmental change. For example, industrial tuna fleets (temporarily) surmount the challenge posed by the decline in tuna stocks through the use of a new technique to intensify extraction: man-made fish aggregating devices (FADs) (Campling). This victory is temporary because fishing on FADs increases the catch of juveniles, which undermines the tuna industry's ecological basis of production.

Exploring Consumption

Exploring consumption links culture, major market dynamics and production systems (Mintz 1985; Fine and Leopold 1993). Only a few studies investigate consumption in fisheries systems. For example, Kurlansky (1998) details how cod has been consumed through history. In the colonial era, cod supported the triangular trade among Europe, North America and the Caribbean by feeding slaves in plantations and labourers aboard vessels. In more contemporary times, rapid growth in consumer demand for cod contributed to species collapse. Bestor (2000, 2001) investigates how a very specific fish cuisine from Japan – sushi – has become a global culinary fashion, with impacts on the practice of producing fish and getting them to newly globalized markets.

Several papers in this issue highlight how consumption trends, including the emergence of 'new' consumer markets, come to bear on seafood production. But markets do not emerge magically; instead, firms may 'create' consumer tastes. For example, through marketing and promotion, Japanese firms have transformed bluefin tuna into the highest-value fish product on the planet (Bergin and Haward 1996; Bestor 2004). Ecolabels have become a dynamic part in several fisheries systems and derive their influence from perceived consumer concerns. Foley and Ponte identify that seafood ecolabel 'consumers' are primarily branded firms, food-service chains and retailers in the global North. Foley investigates the impact of this demand on relationships among fishing and processing firms. Ponte explores how the Marine Stewardship Council, the dominant fisheries ecolabel, *produces* the market for 'sustainable fish'.

The Politics of Fisheries and Development in the 'Global South'

Localized systems of fisheries production are critical to food security and livelihoods in coastal communities the world over, and especially in developing countries (Béné et al. 2007).[12] For example, tens of millions of people rely on the Mekong River's freshwater fisheries for animal protein (Sneddon and Fox, see also Bush 2004). At the same time, '80 per cent of world production of fish and fishery products takes place in developing countries' (Committee on Fisheries 2010, 2). The two issues are related: Alder and Sumaila (2004) argue that the growth of fish exports from the global South to markets in the North can undermine food security in developing states.

International organizations concerned with fisheries-related development collate statistics on fisheries consumption trends, yet generalized data obscure the highly variable political and social processes by which fish are connected to (and/or disconnected from) local socio-economies in coastal communities. For example, a new processing plant in Papua New Guinea introduced tuna to the local socio-economy. While this might appear to improve food security and protein availability, some coastal residents lament that they only have access to damaged and/or inferior tuna products that have been rejected for export markets because of their poor quality. Many residents prefer to eat fresh reef fish caught locally, but argue that industrial tuna fisheries have undermined these food sources (Havice and Reed).

In general, the politics of unequal fisheries development and resource use are not well understood. A notable exception is Kerala where social and political struggles over fisheries and development have been well documented and explored (Kurien 1980; Kurien and Thankappan Achari 1988; Hapke 2001; Subramanian 2009). In this issue, Sinha further enriches our understanding of fisheries development in this region by describing the historical and global dynamics that together constitute a 'transnational development regime' that has shaped contemporary struggles in the sector. He details the complex assemblage of social forces interacting through these fisheries and how they have profoundly influenced the trajectory of transnational fishworkers' movements.

Foreign fishing and fish processing firms are often active in the waters of poorer coastal states rich in fisheries resources (see e.g. Kaczynski and Fluharty 2002; Melber 2003). Associated uneven power relations can generate inequitable distribution of fisheries-related wealth and social conflict, dynamics that have been well explored in a domestic context in the racially segregated fisheries of South Africa (van Sittert 2003; Raakjær-Nielsen and Hara 2006; Ponte and van Sittert 2007; Hara 2009). Ponte's study in this issue highlights one of the ways that fisheries in the global South are excluded from global markets. Producers in the South that cannot comply with the technical demands of MSC certification and the scientific and management infrastructure that it requires cannot access markets for sustainable seafood (which are primarily in the North). Havice and Reed explore the Papua New Guinea state's effort to overcome exclusion from the global commodity chain in canned tuna. To redress the inequitable distribution of revenue from tuna resources caught off its shores, the PNG state has exerted its sovereignty over tuna in its national waters to compel some firms to invest in the domestic manufacture of canned tuna.

In sum, market dynamics and competition in fisheries production–consumption systems demonstrate how capitalist fishing and processing firms constitute relations of power as they

[12] Fisheries and development has been a long-standing concern of the Food and Agricultural Organization. There is also a considerable academic literature on issues to do with fisheries development *policy*, including in journals such as *MAST* and *Marine Policy*.

go about their business. In the next section, we look at relations of exploitation on fishing boats, in processing factories and elsewhere, as well as moments of resistance against such exploitation.

LABOUR, RELATIONS OF EXPLOITATION AND RESISTANCE

As with the other themes we explore in this issue, the dynamics and development of class tensions and forms of social differentiation by gender, race, ethnicity and religion resonate in agrarian political economy, but have their own particular expression and analytical challenges in fisheries. Most existing critical political economy analyses of fisheries are focussed on these topics, as noted in the introduction to this paper. In particular, sociologists and anthropologists have recognized that the management of fisheries is a social and not just a technical question. Research on traditional fishing tenure systems (Durrenberger and Pálsson 1987; Acheson 2003) and fishers ecological knowledge (Neis and Felt 2000; Daw 2008) has analysed, for example, the interaction and frequent conflict between fishers' social organization and state fisheries science and management.[13] This special issue builds on critical work that has drawn attention to social differentiation within fishing communities (and the societies of which they are a part). Labour exploitation, injury and resistance are analysed in relation to fishing crew and boat owners, fish processing, and fishers' local and transnational movements in the papers by Howard, Howse et al. and Sinha. The overview of this theme that follows draws on lessons from these and other contributions in this volume.

Labour and Social Differentiation in Fisheries

A holistic understanding of fisheries must include an understanding of the labour process that underpins the creation of value across all sectors of the fishing industry. The labour process forms the basis for how fishers and fish processing workers experience and understand the industries in which they work. Creating value from seafood while competing in global or regional markets has generated a diversity of social relations to production that evolve over time. In Howard's paper, we see how individuals may begin work in the fishing industry as hired or kin-based fishing crew. If they earn enough to buy their own vessel (or inherit one), they shift to owner-operator, probably still actively working the vessel with either hired hands or kin as crew. Eventually, a boat owner may cease working on the boat and hire skippers and crew, and if very (economically) successful, buy multiple boats. The balance of these relations within fishing fleets can also shift over time, for example, from predominantly kin-based owner-operators to a ownership of multiple boats and reliance on hired crew and skippers. Davis (1996) contrasts these different dynamics as 'livelihood' and 'accumulation' harvesting. The papers in this issue show how many 'livelihood' harvesters also feel (and must be able to cope with) accumulation pressures.

While relatively harmonious 'crew or team production' (St Martin 2007, 538) may be found on some boats, the pressures of operating commercially and selling seafood commodities in the marketplace can mean that 'the relations between skipper and crew at sea become a key site of struggle' (Menzies 2002, 20). Under such pressures, owners may try to increase power over the boat and the labour process on board as they try to produce maximum value with their

[13] Yet some work uncritically focuses on 'community' both as subject and solution for troubled fisheries (e.g. St Martin et al. 2007), which has the potential to gloss over class dynamics and/or reinforce existing power relations (Agrawal and Gibson 1999).

operations and appropriate a greater portion of the surplus (Howard), or they may cope by increasing their own self-exploitation (Marks). In the first case, crew may begin to identify generically as 'crew', rather than crew of a particular boat (Howard 2012b)

Fishing share systems are a common method for distributing surplus among boat owners and crew (ILO 2003), and have some significant parallels to sharecropping in agriculture (Byres 1983).[14] In a fishing share system, crew are remunerated with an agreed portion of the value of the catch. Researchers that argue that fisheries with a share system are not capitalist assume that the share system is not exploitative of labour (St Martin 2007). Howard contests this argument by showing that boat owners *are* able to use fishing share systems to appropriate surplus value produced by boat crew. She hypothesizes that, for owners, the present Scottish share system functions more like a total budget for crew wages. For many Scottish crew, the share system is experienced as a casual and variable wage. More recently, because of deepening market pressures and changes in boat ownership, there has been a shift to a formal wage system. This shift has been facilitated by agencies that recruit crew in the Philippines and elsewhere and whose main business is supplying labour to the global shipping industry. Recruited Filipino crew do not have the expectation of participating in share systems, their wages are usually paid into Philippine bank accounts, and they are employed under conditions inferior to those of Scottish crew.

Distribution of surplus within the family is common in many fisheries of varying scales. Marks describes the sacrifices that Louisiana shrimping families make to stay afloat as global market pressures deepen in their sector. The 'boat eats first' from the family table, which, in tough economic times, strains intra-household and historically embedded inter-firm relations.

Studies of gender and race in fisheries are well advanced (e.g. Nadel-Klein and Davis 1988; Cole 1991; Muszyńska 1996; Neis et al. 2005; Hara 2009). No paper in this issue takes gender or race as its central analytic. However, several papers highlight women's major role, including in household production and fish selling in smaller-scale and artisanal fisheries in India, Papua New Guinea and the United States (Sinha; Havice and Reed; Marks, respectively). The papers also discuss women working in industrial commodity chains. Howse et al. highlight the shared gendered vulnerabilities of rural and seasonal processing workers in Canada and South Africa, all of whom are women, and in the latter, also racialized vulnerabilities. Wages are particularly low among these fish processing women workers and job insecurity is very high. The painful choices that women confront in their roles as workers in fisheries systems are highlighted in Havice and Reed's paper, where some women must choose among a flexible but uncertain livelihood selling reef fish (which appears to be in decline), the difficult and very unsafe practice of trading sex for low-quality salt fish, and the new option of highly disciplined low-waged labour in fish-processing plants.

Competitive market pressures do not just have the potential to change social relations; they can result in the disease, injury and even death of fishers and processing workers. Work on fishing boats is dangerous and regulating worker safety on vessels is challenging (Power 2008; Windle et al. 2008). Worker safety and health issues extend through to the fish processing sector. Howse et al. show that many workers are made ill through processing shellfish, but that this occupational disease is treated as an 'externality' to which companies, health services and government compensation programmes do not attend. They also demonstrate

[14] A major difference between the two is that share systems in fisheries are organized around access to a boat while sharecropping in agriculture is organized around access to land. The significance of this difference is worthy of comparative investigation.

that fish worker precarity is intensified by the perishability of fisheries products and the seasonal nature of fisheries production (on account of regulation, and/or fish migrations or lifecycles). These cycles create either short-term gluts of fish where workers are under pressure to process high volumes, or extended periods of time where there are not fish to be processed and thus no employment opportunities. Gendered and racialized employment practices and access to health care and injury compensation (or lack thereof) further exacerbate these problems.

Processing industry managers argue that in the context of competitive world market conditions, the cost of implementing measures to improve working conditions would undermine commercial viability (Havice and Reed; Howse et al.). The precarity associated with fluctuating and globally competitive markets results in processing workers whose every breath 'feels like you took a knife and rammed it right though ya' (Howse et al.), and trawler skippers lamenting 'there is not a year that goes by, I don't think, without someone I know or know of killed at the job' (Howard).

Fishers' Organization and Resistance

The tensions that generate social differentiation within societies and fisheries systems can also result in new forms of social organization and resistance. Several papers in this issue touch on fishers' and fish processors' organizations in India (Sinha), Canada (Foley; Howse et al.), the USA (Marks), PNG (Havice and Reed) and South Africa (Howse et al.). De Alessi reveals how Maori in New Zealand organized around their indigenous identity and waged large-scale political protests, in part, to reclaim access to coastal fisheries resources that their ancestors extracted before and during the colonial era.

Sinha makes a critical intervention into debates on transnational agrarian movements (Borras et al. 2008). He calls for a long and rich history of the formation of movements, attention to how the unique features of fisheries shape them, and argues that researchers on transnational social movements need to emphasize complexity within these networks. His analysis reveals that Keralan fishers' transnational contacts were (and continue to be) constitutive of the diversity of forms of fisheries and worker organization in multiple historical periods. Today, the formal international networks of fishworkers that emerged in part from the Keralan movement remain influential in international fisheries politics. They include the International Coalition in Support of Fishworkers (icsf.net) and the transnational network of fishers' organizations that was established at the first meeting of the World Forum of Fish Harvesters and Fish Workers in 1997 (worldfisherforum.org). However, tensions related to participants' different social and geographical positions in the global fishing industry partially fractured the movement.

As economic pressures in fisheries mount, owners and crew may develop divergent interests. Scotland is a case in point: low waged Filipino crew found that the existing regional 'fishermen's organizations' – funded and run primarily in the interests of boat owners – did not represent them. Instead, they turned to the International Transport Workers' Federation – which primarily represents seafarers in the offshore oil and shipping industry – for assistance (Howard). Class tensions expressed through activity in trade unions are present in Newfoundland, where each year unionized fishing interests[15] bargain with a processing association to set the prices that processors will pay for catch. These negotiations have recently resulted in industrial action by both fishers and processing firms (Foley).

[15] Newfoundland fishers are represented by the Fish Food and Allied Workers' union, part of the national Canadian Auto Workers union.

Fish processing generally takes place in large, factory-like workplaces using waged labour. Yet even in these more obvious sites of union organizing, the existing structure of the fishing industry presents substantial obstacles to worker efforts to improve wages and conditions. The frequently seasonal and often volatile nature of catches means that employers usually seek to keep employment relations as casual and flexible as possible. Processing plants may also be located in areas with high rural unemployment, giving plant operators more political power over workers fearful of losing their jobs (Howse et al.). Despite these general obstacles in the industry, workers resist. For example, Havice and Reed describe two substantial labour strikes even in the very difficult circumstance of fish processing in Papua New Guinea.

Class tensions have long been a feature of work at sea. For example, it is believed that the term 'strike' originated with seafarers in the eighteenth century, who realized that the captain was powerless to sail the ship if they 'struck' the sails (Rediker 1989, 110). Yet strikes are less common on *fishing* boats where share-based payment and kin-based employment has often promoted a unity of interest among all fishing crew (Lummis 1985). Strikes and union organization are more common in industrial fisheries where some processing is done on board, such as in Norway, Alaska, and Argentina. Many of these unions are part of the Fisheries section of the International Transport Workers' Federation (ITF), and the International Union of Foodworkers (which includes unions representing workers in fish processing factories) have recently launched a joint fisheries organizing project with the goal of improving the representation of fisheries workers on boats and in processing plants around the world (ITF/IUF 2011). The ITF has also been involved in a long effort to draft and then to have states ratify the International Labour Organization's Work in Fishing Convention 188 in an effort to improve working conditions on fishing boats (ITF 2011).

Owners and managers of boats and fishing companies also organize to improve their position. In the European tuna fishing industry, industry associations enable firms to co-operate and use their collective bargaining power in relations with national and regional governments, including in fisheries access negotiations (Campling). Foley describes efforts by firms to collaboratively resist attempts by unions and inshore fishers to maintain or improve prices. In other circumstances, employers can make alliances with fishers' unions, as in the efforts by Indian fishers and boat owners to exclude more capitalized foreign fishing vessels (Sinha).

NGOs play a substantial role in advocacy on various aspects of the fishing industry, primarily environmental (e.g. Greenpeace, Oceana, WWF). Some, such as the Environmental Justice Foundation and the Coalition for Fair Fisheries Arrangements, focus on human and labour rights in fisheries (EJF 2010) and interactions between industrial fleets and artisanal fishers (Gorez 2009). Sneddon and Fox highlight the role of some NGOs in contesting hydropower projects because of the impact they will have on fisheries and fisheries-based livelihoods. Sinha reveals contentious politics around NGO activity by describing how NGOs and 'civil society' involved in fisheries debates in India strategically positioned themselves in the development regime to advantage one group of fishers over another. Overall, NGOs of varying remits and types are very active in fisheries issues globally and across scales (from industrial to artisanal), and their role in fisheries systems is worthy of further investigation.

In sum, creating value from seafood in the context of world market competition has generated a diversity of social relations to production that evolve over time. Many of these are familiar to debates in agrarian political economy, such as gendered and racialized labour relations, class politics, exploitative working conditions and modes of resistance. The papers in this issue draw out the specificity of these relations in fisheries systems, where for example, the share system used to distribute surplus among owners and crew is strained by market dynamics, and the seasonality of fishing seasons heightens worker precarity. In the next section, we look at several dimensions of resource access in fisheries systems in which social–property relations take peculiar forms.

RESOURCE ACCESS AND STATES

In examining the theme of resource access and the role of states in fisheries, we step back from market dynamics and relations of production to investigate the wider politics and mechanisms that regulate fishing activities and relationships. This theme connects the unique features of fisheries resources to the mechanisms that govern productive relations in fisheries. Three papers centre on this theme, each conceptualizing 'access' differently: from historical struggles by Maori over fisheries access rights in New Zealand (De Alessi), to the use of resource sovereignty by the Papua New Guinea state in an attempt to lure fish processing investment onshore (Havice and Reed), and to the commercial struggles around ecolabelling among fishing and processing firms seeking to secure resource access in Canada (Foley). Here, we draw on these and other papers in this issue to summarize the dynamics by which fishers and fishing-related firms secure resource access.

Much academic and policy work on fisheries access relations reduces 'access' to fisheries management and its failure to control fishing pressures. Liberal institutionalists conceive of resource access, and its environmental outcomes, as determined by weak property rights regimes and distorted market signals in fisheries systems: technical components of fisheries systems that can be controlled by centralized regulatory policy frames (e.g. Webster 2009; DeSombre and Barkin 2011). This approach fetishizes positivist rule-making and models of efficient modes of resource extraction; a trend worth noting because the 'academic' and 'applied' intersect around management – models inform fisheries management in practice.

Assessments of the policy and practice of resource access in this volume suggest that, whether conceptualized around capturing 'maximum sustainable yield', ecosystem-based management or using fisheries resources for socio-economic development, management (and related social–property relations) is a political process that is inextricably bound up with processes of capitalist accumulation. The papers in this issue reveal that fisheries management (whether coordinated by the state or non-state entities) is inevitably characterized by unintended (and often uneven) consequences and outcomes. The agrarian political economy tradition enables an assessment of the politics of resource access that 'focuses on the issues of *who* does (and who does not) get to use *what*, in *what ways*, and *when* (that is, in what circumstances)' (Neale 1998, 48, italics in original). Following from Ribot and Peluso (2003), resource access is a bundle of powers that enable actors to gain, control and maintain access to and benefit from resources.

Within this framing, 'access' is not only a site of political contestation narrowly organized around fisheries management, it is defined by many moving parts (such as access to capital, indigenous identity, or market access) that dictate fisheries use patterns and their socio-economic and ecological outcomes. Several contributions in this issue link this general tradition of investigating resource access from a political-economy perspective to the unique challenges that emerge when fisheries are the resources in question. They build on earlier work that illuminates how defining, securing and using fishing access is a political endeavour shaped by geographical, historical, social, economic and ecological factors (Sinclair 1983; Mansfield 2004b; Havice and Campling 2010).

An analytical thread that links several papers is that the presence of a marine resource neither indicates, nor determines, the relations of production around that resource. Instead, access relations serve to link, or de-link, the geography and economy of fisheries production, while 'managers' act to control and distribute fisheries access. For example, European fishing firms have expanded into fishing frontiers in the Western Indian Ocean to secure access to raw materials that supply not African, but European markets (Campling). In Papua New Guinea, the state regulates fishing licences in the hope of creating domestic fisheries-based economic opportunities (primarily jobs in tuna processing plants). The state, foreign firms and coastal residents have competing visions of what it means to derive 'benefit' from resource access. This dynamic generates a significant wedge between the state's intended effort to use its control over tuna access to generate jobs and the resulting local-level industrial change experienced by coastal residents (Havice and Reed).

In New Zealand, Maori used identity – more specifically, indigenous identify – to acquire and control fishing rights from which they were historically excluded. De Alessi explores how Maori entry into fishing was moderated by a combination of Maori interactions with colonial and post-colonial governments. This analysis reveals that instituting private property rights (in the form of individual transferable quotas, ITQs) is not simply about maximizing efficiency in fisheries production system, but is instead complicated by political, economic and social relations. On the political, the state acts as an assemblage of interests (not simply a functional representation of capital) and allocates ITQs to Maori historically excluded from fishing rights. On the economic, control over ITQs enables *some* Maori to overcome the barriers that have prevented them from participating in the fishing industry (e.g. acquiring the capital to buy or lease more quota and/or vessels). But, ITQs also produce social differentiation between Maori who gain and control access to fishing quota and those who do not.

'Sustainability' certification schemes are emerging as sites of struggle over resource access. Two papers investigate the political economy of the Marine Stewardship Council. As some retailers commit to sourcing only 'sustainable' seafood, Foley shows that MSC certification has emerged as a new tool for fisheries access. Firms that hold or possess the MSC 'certificate' (in this case, shrimp processing firms) have gained new power in some fisheries commodity chains. Foley's story reveals that the inability to draw property boundaries through fisheries stocks and fleets means that defining who is an MSC client and to whom the certificate applies influences resource access and production dynamics. Ponte illuminates another subtle dimension of fisheries resource access related to 'sustainability' certification: certification is, in and of itself, a market. To gain and secure dominance as *the* fisheries ecolabel in the seafood market, the Marine Stewardship Council, as an organization, must find a balance between accumulation imperatives (certifying as great a volume and value of fish products as possible) and legitimation (demonstrating that certification generates ecological improvement and extending the certification to small-scale and developing country fisheries). He argues that the balance has so far tipped towards accumulation and that the MSC has struggled to legitimize its certification processes in terms of environmental and social improvements.

Access to capital also plays an important role in mediating fisheries access. In De Alessi's example, government redistribution of fishing quota to indigenous groups enabled some Maori to break the capital barrier to entry; valuable quota was (and is) used as an asset to foment productive investment. In Howard and Marks' examples, some crew in Scotland and Louisiana historically accumulated capital to fund upward mobility into vessel ownership. However, in both cases, recently intensified world market competition has constrained possibilities for upward mobility. To cope with competitive dynamics vessels are increasingly capitalized; this makes it more difficult for crew to accumulate enough money to buy a boat to directly access fish resources.

The History and Diversity of Social–Property Relations in Capture Fisheries

That fisheries resources are mobile and offshore, and at times straddling or migrating between the jurisdictions of multiple states, creates well documented institutional challenges associated with designating fisheries ownership and use rights (Ostrom 1990; Ostrom et al. 1999). Social–property relations in fisheries are not all the same. Many different parties can control fisheries property rights (in different places and times), including: feudal lords, states, indigenous groups, individuals and/or capitalist firms, to name a few. In some cases, the *lack* of property rights, the condition of an 'open access' fishery, has considerable impact on patterns of resource use. As is the case with terrestrial resources, those with the privilege of controlling property rights over fisheries resources hold social power over production choices, though not necessarily its ecological and social consequences.

While states remain a central – though not the only – institution that recognizes, determines and allocates fishing rights, the politics and practice of exercising rights is varied across history and fisheries. Longo and Clark reveal how bluefin tuna property relations have shifted over time. In the first documented concessions, the Norman kings of Sicily offered rights to specific bluefin fishery trap systems (*tonnare*) to monasteries and Bishops (late 1000s and early 1100s), and later to nobility (early 1200s). During this time, 'rights' to bluefin tuna were oriented around ownership over the traps in which they were caught. As the power of the kings of Sicily diminished and the transition from feudalism to capitalism took hold (1600s), private individuals came to acquire the bluefin fishery trap systems and Genovese bankers invested. Industrial bluefin fishing rendered the trap system non-competitive in the contemporary capitalist era. The trap fishery collapsed and bluefin property relations are now governed by states. Today, fishing rights are negotiated by an intergovernmental organization that determines a total allowable catch and distributes the catch to the member states, which in turn, allocate fishing licences (see below).

Howard also traces the historical trajectory of fishing rights in Scotland. In the 1500s fishers were initially serfs who gave the landlord a portion of catch in exchange for the use of a boat, land and house; a kind of sharecropping that straddled land and sea. As private land rights were instituted in the 1800s, landlords cleared serfs from their land and later employed the newly formed rural proletariat as fishers. Historically, the right to fish and to earn revenues from the activity has been more closely linked to control over fishing capital and infrastructure, such as ports, boats and transport, rather than the ability to access or purchase quotas regulated by the state.

In New Zealand, the colonial state partially recognized Maori fishing rights through the Treaty of Waitangi in 1840. In subsequent years, Maori were not participants in the industrialization of New Zealand fisheries (De Alessi). However, from the late 1980s onwards, the post-colonial state created 'rights' to fish. When it implemented ITQs and distributed them to established fishing interests, Maori evoked the Treaty of Waitangi and their 'tribal' identity to secure an ITQ allocation and began undertaking capitalist fishing endeavours. In short, ITQs became an entry point for some Maori to appeal to the post-colonial state and negotiate a claim in the distribution of ITQs, demonstrating that fishing rights, and control over them, is a political and economic relation that entails both access and exclusion. Rights-based fisheries programmes and allocation of fishing access through ITQ systems is the dominant paradigm in fisheries management today. The privatization of the oceans *qua* property rights over mobile fisheries resources is an attempt to avoid Hardin's seemingly inescapable 'tragedy of the commons' (Hannesson 2004), though little work has assessed the socio-economic outcomes of the privatization of aquatic resources.

We would be remiss to overlook the significance of those fisheries historically or currently characterized as operating under 'open access' regimes without property rights. The oceans were open access until coastal and island states declared economic sovereignty over their 200 nautical mile exclusive economic zones (EEZs) from the mid-1970s onwards. This had a lasting effect on the formulation of the international division of labour of global fisheries production systems. For example, when oceans were open access, access to capital was the major barrier to entering industrial fisheries in international waters. The lack of domestic capital meant that interests in countries such as Papua New Guinea with rich fisheries, but a lack of capital needed to extract them, were effectively excluded from fisheries commodity chains. Since declaring its EEZ, the Papua New Guinea state has incrementally asserted sovereignty over tuna fisheries to strengthen its position in the international division of labour in the tuna industry (Havice and Reed).

As noted above, Marine Stewardship Council certification has created a new dynamic in fisheries property relations, operating as a *de facto* property right over fish for those firms that 'hold' or 'own' the certification. Since the state *requires* that a component of the Canadian shrimp fleet offloads its catch domestically for processing, and shrimp buyers are increasingly mandating that product is certified as 'sustainable', the firms that hold the certificate are able to secure access to resources, and exert influence over the terms by which they procure the shrimp products (Foley).

States and Fisheries Production

States are front and centre in many of the analyses in this issue. We attribute this, at least in part, to the unique challenges of governing resources that are mobile and to state interests in fisheries-related revenue. Several papers reveal the conflicting demands on the state to mediate diverse fisheries interests. This highlights how some states use (or avoid using) regulatory and enforcement capacity to generate outcomes that privilege particular players in fisheries systems.

State strategies to control and develop fisheries resources vary. In India, state agencies identified fisheries resources as 'under-utilized' and developed programmes to draw fisheries resources into broader economic and social development objectives.[16] Sinha demonstrates that, beginning in the 1940s, the Indian state identified fisheries development in Kerala as a priority for generating revenue (including foreign exchange) and increasing fishers' income and protein intake. To generate fisheries development, state agencies in India secured financing through foreign aid and created institutions to provide economic, scientific, technical and marketing support. This spawned both a new fisheries regime and tensions between productivist and welfarist, foreign and national, national and regional, and industrial and small-scale interests in fisheries. In New Zealand, the colonial state promoted the development of industrial-scale fishing through state subsidies, but Maori were excluded from subsidy programmes, and as a result, from participating in, and benefiting from large-scale fishing activities (De Alessi). In the Mekong, the geopolitics of a shared/ trans-boundary river system in both pre– and post–Cold War eras influence how the use of river resources are prioritized in relation to regional economic development (Sneddon and Fox).

States also play a role in the regulation of capital–labour relations in fisheries production. In detailing working conditions in fish processing facilities in Canada and in South Africa, Howse et al. show that by neither preventing workplace illness or injury nor providing adequate care

[16] Increasingly, fisheries are 'over-utilized', generating new dynamics around state management (see conclusion).

for injured and ill workers, the state can be seen as either failing to regulate the fish processing industry or as regulating labour to reduce costs for capital. In Newfoundland and Labrador, the state mediates the relations between labour and capital by facilitating collective bargaining agreements between fishers and fish processors over the price of catches (Foley). It also plays a significant, but under-acknowledged role in facilitating 'private' sustainability certification because the MSC certification requires having a strong state-led fisheries management plan in place.

Science and Management

Last, but far from least, is the role that science plays in informing politically motivated resource management decisions, and in turn, fisheries access. Much work in the agrarian studies tradition has explored the role of science in agricultural production and agricultural policy, focusing, for example on the social, environmental and economic effects of biotechnology in food production (e.g. Kloppenburg 2004; Weis 2007). This includes a discussion of how science, when applied in food production, impacts control and access for agricultural production and consumption practices. Such work critically analyses how new forms of science empower various interests groups and, in the process, generate policy discourses that determine which actors produce natural resources and which resource-based food products are available to them (e.g. Bernstein and Woodhouse 2001; Scoones 2006). More generally, this *Journal* has recently identified the need to look more closely at science as part of the productive forces (Bernstein 2010b; Wield et al. 2010). Certainly, as climate change has begun to alter the environment, the role of science in quantifying, predicting, and informing policy decisions has come front and centre.

These general trends apply directly to fisheries systems where scientific models of fish stocks, ecosystem dynamics and the impact of fishing effort on biological and ecological dynamics in the water provide the data upon which fisheries management decisions are made. But there is relatively little work on the political economy of the formation, uptake and application of science in fisheries systems, despite the critical role science plays in management and related resource access issues (notable exceptions include Finlayson 1994; Bavington 2010). While contributions to this issue engage the scientific underpinnings of resource access decisions, none take fisheries science as its analytical vantage point. We highlight that further investigation of the role of science in political economy analyses of fisheries production and management is a fruitful research direction for scholars working on fisheries.

That said, fisheries science makes significant cameos throughout this issue, hinting at the diversity of ways in which fisheries science is created, deployed, and of course, politicized. We see that in some cases fisheries science is ignored, while in other cases it is strategically deployed as states negotiate the terms of fisheries management. On the former, the scientific management of highly migratory species such as tuna requires inter-state co-operation, but often, management decisions reflect commercial interests rather than scientific recommendations (Campling; Longo and Clark). On the latter, regional planning bodies in the trans-boundary Mekong Basin generated science that under-emphasized the ecological diversity and productivity of freshwater fisheries. Meanwhile global eco-hydrology experts stressed the productive potential of capturing the river's flow to provide hydropower, irrigation and flood control. In short, governments created a story of fisheries science and used it to argue that the ecological processes and conditions that produce fish in the river delta are unable to lift the region out of poverty, while the science behind dam development was deployed to argue that hydropower could (Sneddon and Fox).

Science remains important outside of traditional, state-led fisheries management in forms as diverse as legitimizing private certification and supporting policy demands of fisherworkers' social movements. The Marine Stewardship Council relies on scientific data on stock status and bycatch levels to determine if it can grant a product the 'sustainable' seal of approval. Since several major buyers now require MSC or compliance with other environmental standards, scientific approval has become a critical step in securing market access. Further, to lend itself legitimacy, the MSC has a technical advisory board whose membership is more than half comprised of fisheries assessment and/or management scientists charged with verifying that the MSC methodology does indeed determine which stocks are 'sustainable' (Ponte). In India, fishworker activists generated cutting-edge scientific and social scientific research on fisheries decline and the social conditions of production to ground their demands for improved working conditions. The movement created its own institutional complex of expertise, generating hubs for fisheries research that could in turn be folded into government policy forums (Sinha).

In sum, resource access is a central component to understanding fisheries systems. Defining and allocating social–property relations in fisheries is a complex and contested process, not least because fish move, blurring traditional conceptions of property. Fisheries access is moderated by state managers, informed by science and is inextricably bound up with processes of capitalist accumulation. Counter to accounts that reduce fisheries access to 'management' alone, a political economy reading reveals access as constituted by social relations that reflect and refract specific histories, geographies, market dynamics and relations of inclusion/exclusion.

CONCLUSION AND FUTURE RESEARCH

Fish are a renewable but exhaustible resource and the world's last hunted mass commodity. Their mobile nature is associated with peculiar social–property relations. The diversity of ways in which fish are captured and consumed or brought to market results in equally diverse forms of social and economic relations, mediated in various ways through capitalist relations of exploitation and commodification. As a whole, capture fisheries systems are experiencing rapid socio-ecological change. We have sought to analyse distinct fisheries side by side and within the broader political economy of which they form a part. Papers in this issue build off of an existing corpus of social science literature and engage several themes familiar to readers of this *Journal*, including: (uneven) market dynamics and competition, relations of exploitation and resistance, and historical and contemporary forms of resource access in both the global North and South. The analysis in this introductory paper highlights some of the ways in which the unique combination of characteristics associated with fisheries complement and complicate these foundational themes.

The authors' contributions in this issue take readers on a journey far and wide, covering the global oceans (except the Antarctic) and one extraordinary riverine system. While the papers expose the diversity of fishing activities and techniques, they do not capture the socio-ecological range of fishing in its entirety. Geographical, thematic and conceptual gaps in this issue's coverage offer sites and subjects for future research, including, for example, in East Asia, Latin America and the Middle East, and on lake systems.

Papers investigate worker safety in fish processing plants and on board vessels, but working conditions in the sector are surely worthy of more attention. We have drawn a bright line in the sand to devote this special issue exclusively to capture fisheries, but the volume and value of aquaculture continues to grow in significance relative to capture fisheries. Aquaculture has intersections with capture fisheries that are worthy of investigation and explication. In this issue, we saw hints of the ways that capture fisheries and aquaculture interact (e.g. through the market dynamics of competition in the global shrimp industry). Likewise, gender and race are centrally important to understanding social relations in fisheries, but are examined only peripherally here. This issue's content tips in favour of export-orientated fish production, though local fisheries for local consumption are highly significant, particularly in the developing world.

Environmental change is impacting the political economy of fisheries. Increasingly, new social and economic relations are being generated as people cope with 'over-utilization' and scarcity of fish resources. There is need for future research on how themes introduced in this paper (e.g. the capitalist growth imperative and the political economy of resource access) intersect with fisheries extraction and concomitant exhaustion. An emerging body of natural science research is documenting the impact of climate change and ocean acidification on fisheries, but there is very little critical social science work on their effects on fisheries-related development (Allison et al. 2009; Robinson et al. 2010). Fisheries politics are changing as managers implement new tools such as marine protected areas and marine spatial planning. These policy efforts at once offer potential for enhancing fisheries productivity, protecting nature as 'wilderness' (not as a location of production), and generating conflict among commercial fisheries, recreational fishers, and non-fishing industries (e.g. tourism and wave energy).

Fisheries have long had an important place in geopolitics as nations compete for access to fisheries resources and use national fishing interests to gain political influence. Familiar examples include the cod wars (Jóhannesson 2004), Cold War contests for spheres of influence that played out in battles over fishing access rights (Doulman 1986; Havice 2009), and Taiwan's recent efforts to gain influence in regional fisheries management organizations as a stepping stone to official diplomatic recognition (Havice and Campling 2010). Glassman (2011) points out that the geopolitical aspects of global production networks can extend our understanding of the process of capitalist globalization. Fisheries production-consumption systems are ripe for this type of examination, which is only posed to become more significant as melting sea ice and developing aquaculture techniques open new aquatic frontiers.

The majority of capture fisheries are at or beyond full extractive capacity, heightening the need to examine the political economy and ecology of the interests driving decline. This is particularly important since fish are a significant source of food and employment at subsistence and industrial scales. The contributing authors illuminate the forces behind these trends by using a range of methodological approaches and analytical emphases to demonstrate the historical, social and ecological complexity of capture fisheries, and the diverse ways in which capitalism works through the particular environments of the sea in the process of bringing fish commodities to markets. In doing so, they make important inroads to understanding socio-ecological change in fisheries systems. We hope that this special issue encourages future research in this vein.

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